

Development of driving cycle for Sri Lanka

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Abstract

A driving cycle is a speed time profile which represents the driving characteristics of a selected area. This can be of use to legislative and non legislative purposes such as development of emission standard, determination of fuel consumption, etc. Established driving cycles cannot be used in Sri Lanka as the traffic constituents, driver behaviour and road conditions vary from the conditions of those countries. Hence the objective of the research is to develop methodology and standard driving cycle(s) which represent urban and rural conditions in Sri Lanka and to establish emission standards in future based on such cycle(s).

To achieve objectives, it is necessary to collect and analyse the on road speed-time data on selected routes which will represent the driving patterns of urban or rural conditions. Due the restriction of time and resources, our study was limited to urban condition and for light vehicles; a sample driving cycle was developed based on the data collected on the Galle Road, section from Katubedda to Fort using on board method, during different times and days of the week. This study would facilitate the further studies on development of driving cycles for other conditions and this could be of great use for traffic engineering studies and sustainable development.

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Extended Abstract

A driving cycle is a speed time profile which represents the driving characteristics of a selected area. This can be of use to both legislative and non legislative purposes such as development of emission standard and determination of fuel consumption. Established driving cycles such as European driving cycle and United States driving cycles cannot be used in Sri Lankan context as the traffic constituents, driver behaviour and road conditions vary significantly that from the conditions of those countries. Past studies done in Hong Kong and Vietnam have proven this. Hence the objective of the research is to develop a standard driving cycle(s) which represents Sri Lankan urban and rural conditions and to establish emission standards in future based on such cycle(s).

To achieve stated objectives, it is necessary to collect and analyse on road speed-time data on a selected route which represent the driving pattern of the urban or rural area. This should be done paying attention to the route selection. The route for data collection should be selected based on the traffic density and its vehicle composition, population density, topography, land use, climate conditions, road surface conditions, mean length of a trip, number of signalized intersections on the route etc. and data should be collected considering the peaks of the day such as morning, noon and after noon peaks as those times would be critical in the case of traffic condition, driver behaviour and emission levels.

Cycle construction can be based on either Micro trip based Cycle construction, Segment based Cycle construction, Pattern classification Cycle construction or Model Cycle construction. The basic difference in above methods is the selection of random samples. And this may have various advantages and disadvantages for the intended use of the cycle as emission related purposes or traffic engineering studies. In general, a driving cycle developed based on randomly selected on road speed data such that summary statistics of the developed cycle will represent the whole test run population. For this exercise both principal component analysis (PCA) and cluster analysis are used. The basic target parameters used are idle proportion, acceleration proportion, cruise proportion, deceleration proportion, RMS acceleration, average acceleration, average deceleration, average speed and maximum speed. Out of the candidate driving cycles developed, the best will be chosen based on the target parameters and the speed acceleration distribution.

A sample driving cycle was developed based on the speed-time data collected on the Galle Road, section from Katubedda to Fort. This road section was selected considering the parameters stated above, because this would represent a typical urban driving condition in Sri Lanka, hence the developed driving cycle can be more representative. Due to time constraint, an on board method (chase car method has some complexities) was used to collect data using a hand held GPS device. More than 30 samples were collected on this section during different times of the day and different days of the week to represent many prevailing traffic conditions at this urban road section. The driving cycle may vary from vehicle type and thus in our study was limited to light duty vehicles. Cycle construction was done base on the micro trip cycle construction as it is more suitable for emission related studies. To facilitate cycle construction a computer application was developed based on the micro trip cycle construction methodology. This application divides the given set of data into number of micro trips after filtering data for maximum speed and acceleration and it chooses micro trips randomly such that the average speed of the cycle developed is approximately equal to that of the developed candidate driving cycle of 20 minutes. Further, this application lists out the target parameters of each candidate cycle and thus, best cycle could selected manually based on PCA and cluster analysis.

This study was limited for the development driving cycle for an urban condition and light vehicles. This study would facilitate the further studies on this area and development of driving cycles for rural, sub urban and express way condition and this could be of great use for traffic engineering studies, road safety and sustainable development.

Conclusions

Based on the model driving cycle developed for a Urban condition in Sri Lanka the following out puts can be seen. The average speed 21.37km/h, average running speed 25.78km/h, maximum speed 72km/h, average acceleration 2.03km/h/s and average deceleration 2.02km/h/s. And the test population showed acceleration, deceleration and cruising proportions as 27.37%, 23.70% and 48.93% respectively. When compare, these values differ from the existing driving cycles, hence reiterates the continuation of research. This study can be considered as the starting point of driving cycle development for Sri Lanka, which truly represents the local conditions. Forecasting fuel emissions, travel time predictions, setting up emission standard, etc. are among many of the expected out puts from driving cycles.